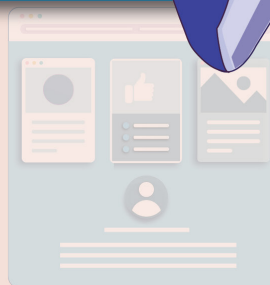




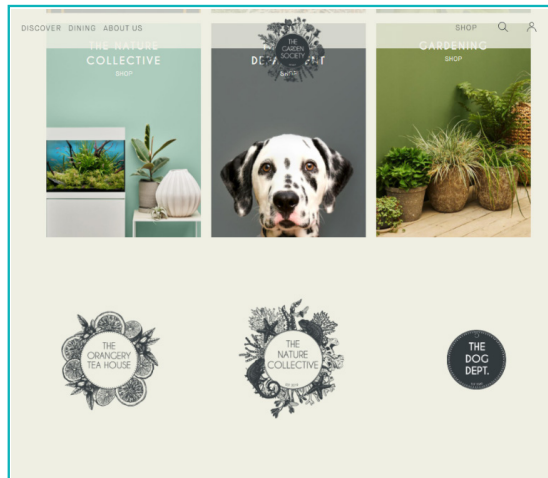
Web Layout Design

05

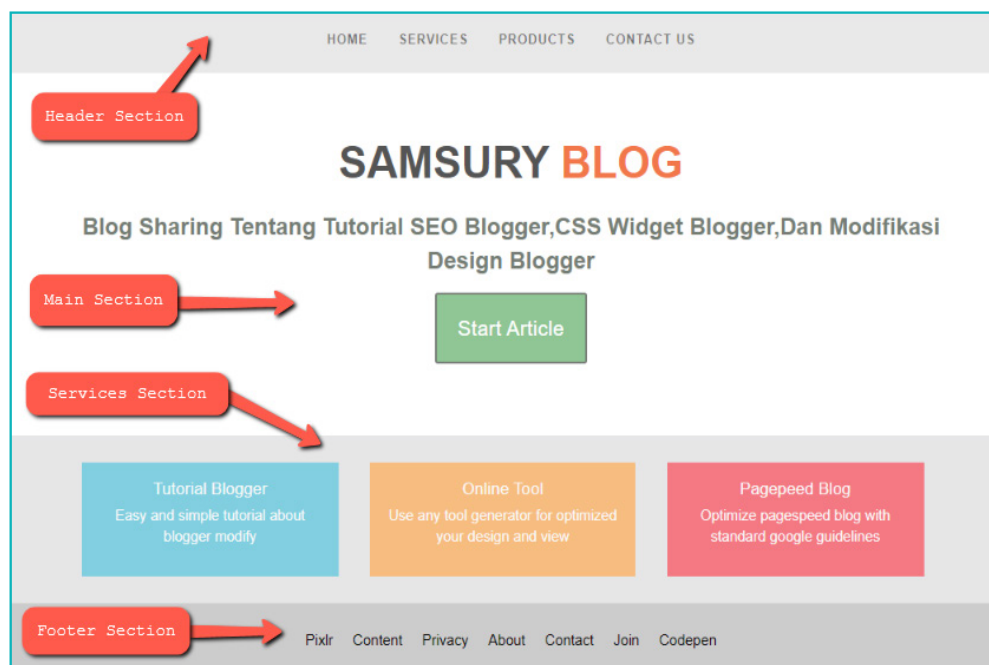
Most websites that we visit look amazing because they are visually attractive and convey messages effectively with better readability.



Which websites do you frequently visit? Have you noticed how beautiful some webpages look? The contents are perfectly aligned, the images are chosen well, and the overall design makes the website shine.



Take a look at the webpage shown below. The site is beautifully designed and categorized into multiple sections (header, main, services and footer). Different colours are used for text and other components.



Importance of a website layout

If we get rid of the web layout and the design, our website will look plain and boring, like the image below. It just contains text and looks dull and unappealing.



Apart from it being colourless, the contents are no longer in their proper position. They are just placed as a list, one below the other. It doesn't look like a website that anybody would want to visit. A website's layout and design breathe life into it.

At this point, it should be clear that it's not only the colour, fonts or images that makes a site look appealing, it's also how all the components are organised within a website.

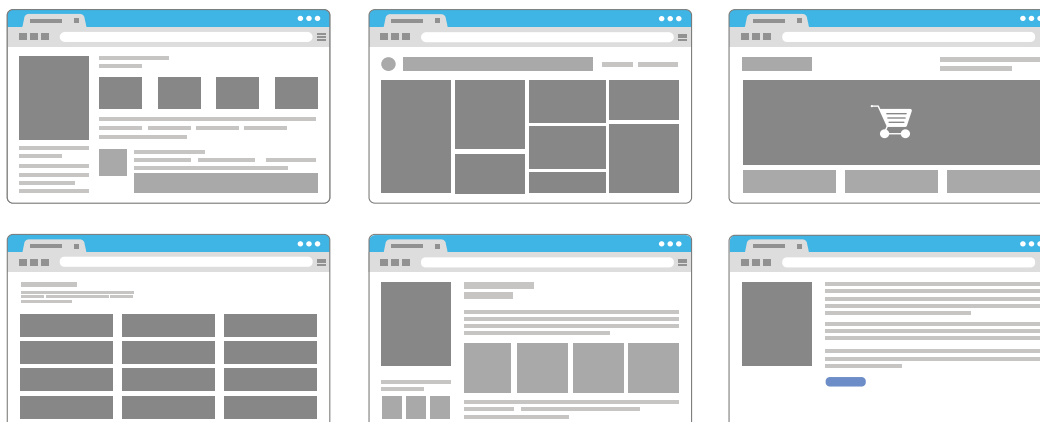
A website's appeal is not limited to the colour, font or images. How the components are arranged and organised is equally important. To design this, we need to learn about website layouts.

Understanding a website layout

A website layout is a pattern or framework that defines a website's structure. Its role is to structure information on a site for the website's owner and its users. It provides clear navigation paths within webpages and puts the important elements in front and centre.

The figure below shows 6 different layouts with differently arranged components. Components are depicted in grey.

There is no fixed format for a layout. It depends on the website contents and how we want them to be displayed.



Before creating a website layout:

- We need to visualise a layout first, and then start to code. This will help us understand how to organise its contents.
- A site may have a header section, main section, products section, or a footer, so the layout should be planned accordingly. We can create a rough layout on paper too.
- Optionally, we can specify sizes for all our components. If we aren't sure, this step can be skipped for now.

Activity

Design a website to provide services for people's pet animals.

Identify website components

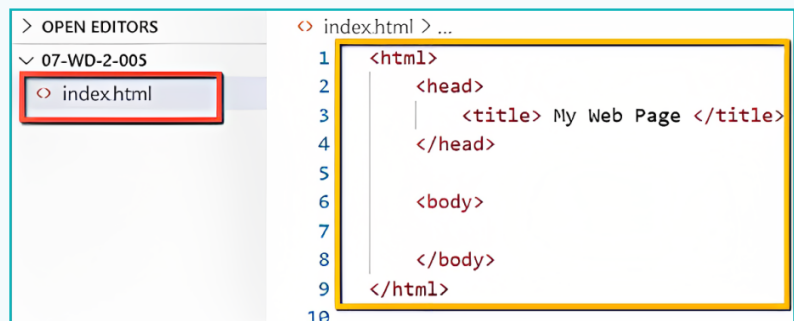
To start, we need to determine which components or sections we want to include. In this activity, we will include the following:

- **Header Section:** The first section with the main heading and title of our website.
- **Services Section:** Contain some basic information about services provided.
- **Dog Walk Timing Section:** Display services specific to pet dogs.
- **Cat Walk Timing Section:** Display services specific to pet cats.
- **Contact Us/Footer Section:** Display contact details for pet owners to get in touch.

Now that we have identified the sections, let's start to create them.

1 Create a folder called "MyWebPage" to save HTML, CSS files and images

- a. Open VS Code under the File menu, click on Open Folder and then select the folder we created.
- b. Create a file index.html and write down the basic HTML structure.



The screenshot shows the Visual Studio Code interface. On the left, the 'EXPLORER' sidebar shows a folder named '07-WD-2-005' with a file named 'index.html' selected. On the right, the 'EDITOR' pane shows the content of 'index.html' with the following code:

```
1 <html>
2   <head>
3     <title> My Web Page </title>
4   </head>
5
6   <body>
7
8   </body>
9 </html>
```

Activity

2 Create separate divisions, or containers, for each section using the <div> tag

The <div> tag is used to define a section or division in an HTML page. We can also use it to group multiple HTML code in a single block.

Note:

As of now, there is no visible output if we run this page on a browser. This is because we haven't added any content inside the div containers.

```
<html>
  <head>
    <title> My Web Page </title>
  </head>

  <body>
    <div> </div>
    <div> </div>
    <div> </div>
    <div> </div>
    <div> </div>
  </body>
```

3 Wrap the containers inside a single div container for easy structuring and layout

```
<body>
  <div>
    <div> </div>
    <div> </div>
    <div> </div>
    <div> </div>
    <div> </div>
  </div>
</body>
```

4 Create an <h1> and <p> tag to add content to the first div container. This will give us a visual representation of the layout at this stage

```
<div>
  <h1> PET'S CARE </h1>
  <p> Your Pet, Our Responsibility</p>
</div>
```

PET'S CARE

Your Pet, Our Responsibility

Activity

5 Repeat Step 4 and add content for the other sections.

```

<div>
  <h1> OUR SERVICES </h1>
  <p> We are a group of animal lovers having
  certification in animal care. We provide various
  services like pet walking, grooming, feeding,
  playing with, and temporary shelter.
  </p>
</div>

<div>
  <h1> Pet Dog Services</h1>
  <p> Walking Services only on Monday, Wednesday
  and Friday - from 4pm to 10pm </p>
  <p> Other Services are provided 5 days a week </
  p>
</div>

<div>
  <h1> Pet Cat Services</h1>
  <p> Walking Services only on Tuesday, Thursday
  and Saturday - from 2pm to 8pm </p>
  <p> Other Services are provided 5 days a week </
  p>
</div>

<div>
  <h1> Get in Touch </h1>
  <h4> Phone Number - XXXXXXXXX</h4>
  <h4> Email - petcare@gmail.com</h4>
</div>

```

OUR SERVICES

We are a group of animal lovers having certification in animal care. We provide various services like pet walking, grooming, feeding, playing with, and temporary shelter.

Pet Dog Services

Walking Services only on Monday, Wednesday and Friday - from 4pm to 10pm

Other Services are provided 5 days a week

Pet Cat Services

Walking Services only on Tuesday, Thursday and Saturday - from 2pm to 8pm

Other Services are provided 5 days a week

Get in Touch

Phone Number - XXXXXXXXX

Email - petcare@gmail.com

6 Identify individual <div> containers

Type Selectors

Here, the selector can be any HTML element.

```
h2
{
  color: red;
}
```

This CSS rule will make all the h2 headings red in color.

ID Selector

This is the most common way of applying a CSS style rule. Here we provide an ID to a HTML element and all CSS styles will be applied only to the HTML element that has the ID. The # symbol is used to indicate a ID.

HTML Element with ID - "my_heading"

```
<h2 id="my_heading"> Hello World </h2>
```

```

CSS style rule for the ID
#my_heading
{
  color: green;
}
```

This CSS rule will make all the h2 headings with id "my_heading" green in color

Class Selector

Here we provide a class to an HTML element and all CSS styles will be applied only to the HTML element that has the Class. The . (dot) symbol is used to indicate a CLASS.

HTML Element with class - "my_class"

```
<h2 class="my_class"> Hello World </h2>
```

CSS style rule for the CLASS

```

.my_class
{
  color: green;
}
```

NOTE: By convention, IDs are unique across the web page, whereas Class can be applied to a single or a group of HTML elements.

- We have 5 divisions and we need to identify each individual div container. How can we identify an individual division element?
- Use CSS Selectors to answer this question.
- CSS Selectors help identify an individual element by assigning it an ID or a CLASS.
- There are 3 types of CSS Selectors: Type Selectors, ID Selectors and Class Selectors. In this session, we will use a Class Selector.

Activity

7 Add a class selector to the code

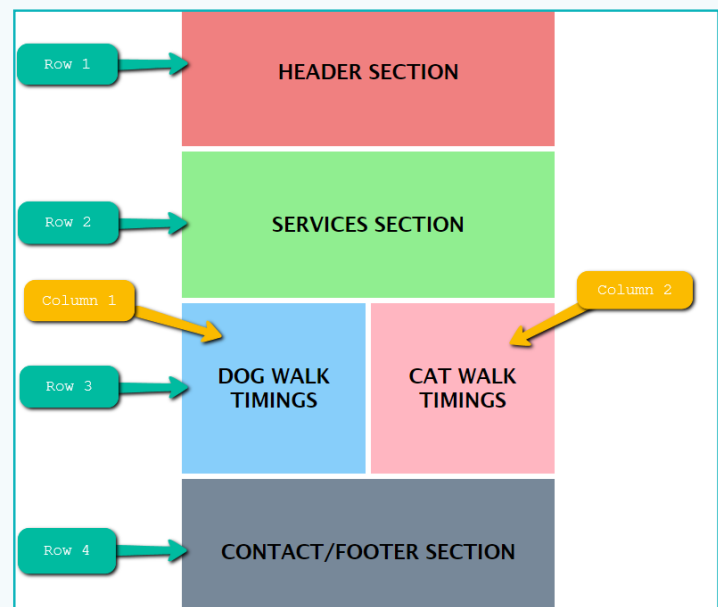
- Assign a class name to an HTML element using the class attribute. We can access the class attribute using the “.” (dot) symbol before the class name.
- We have named all the individual division elements and made them easy to identify.

```
<div class="main">
  <div class="box_1">
    <h1> PET'S CARE </h1>
    <p> Your Pet, Our Responsibility</p>
  </div>
  <div class="box_2">
    <h1> About Me </h1>
    <p> I'm an animal lover and pet owner. I have extensive experience walking, grooming, feeding, playing with, and caring for pets. I've been a volunteer at local animal shelters and have certification in animal care.
  </p>
  </div>
  <div class="box_3">
    <h1> Pet Dog Timing</h1>
    <p> Monday, Wednesday and Friday - from 4pm to 6pm </p>
  </div>
  <div class="box_4">
    <h1> Pet Cat Timing </h1>
    <p> Tuesday, Thursday and Saturday - from 4pm tp 6pm </p>
  </div>
  <div class="box_5">
    <h1> Get in Touch </h1>
    <p> Phone Number - XXXXXXXXXX <br> Email - petcare@gmail.com</p>
  </div>
</div>
```

8 Create a layout using a CSS grid

Create a layout for how each section will appear on the site.

- For this website, we are using a simple layout with 4 rows and 2 columns. For the first, second and fourth row, the 2 columns are merged into a single one.
- To create the above web layout, use a CSS Grid. The CSS Grid Layout or simply “Grid” is a two-dimensional grid-based layout system. Grid makes it easy to create complex layouts.



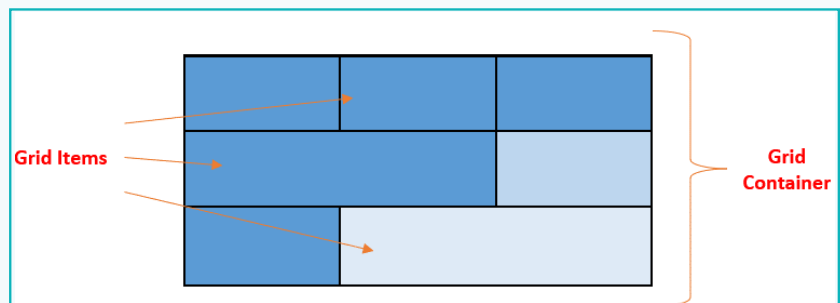
Dummy Web Layout using CSS GRID

Activity

Grid Container and Grid Items:

A **Grid** starts with a **Grid Container**. This contains the elements of the grid.

The image on the right shows a grid container with 3 rows. The 1st row is divided into 3 columns and the 2nd and 3rd row are divided into 2 columns each. We can use this to create custom layouts.



9 Implement a grid on the website

Create a style.css file. Include this file in the index.html page using the link tag as shown below.

```
<head>
  <title> My Web Page </title>

  <link rel="stylesheet" href="style.css">
</head>
```

For the .main class, set the display property to grid. Set the CSS property to display: grid. This converts the main container into a Grid Container and instructs the browser to use the grid layout.

Note:

If we view the page in a browser right now, there will be no change in the output.

```
<head>
  <title> My Web Page </title>
  <style>
    .main {
      display: grid;
    }
  </style>
</head>
```

The next step is to arrange our sections within the Grid Container. We will use Grid Area and Grid Template Area properties. These properties work together to position our items within the grid container.

Activity

10 Name the grid areas

Naming grid areas is the first step to create a grid layout. We will assign a custom name to each of the grid items using the grid-area property.

- In the style.css file, access the division element with class “box_1”. Set the grid-area property to “header”.

```
<div class="box_1">
  <h1> PET'S CARE </h1>
  <p> Your Pet, Our Responsibility</p>
</div>
```



```
.box_1 {
  grid-area: header;
}
```

- Use the same process and add a grid-area name to all the other grid items.

```
.box_2 {
  grid-area: services;
}

.box_3 {
  grid-area: dogs;
}

.box_4 {
  grid-area: cats;
}

.box_5 {
  grid-area: contact;
}
```

11 Arrange components within the grid area

We will use the grid-template-area property to arrange components in the grid. This property specifies an area within the grid layout. We provide a grid area name to fill the areas in the way we want.

- In the .main container, use the grid-template-areas property and arrange sections as shown below.
- This property and values describe a grid with 4 areas. The 1st, 2nd and 3rd row span across 2 columns. We can make an area span across multiple columns by repeating its name in all the grid cells it should cover.

```
.main {
  display: grid;
  grid-gap: 20px;
  grid-template-areas:
    "header header"
    "services services"
    "dogs cats"
    "contact contact"
  ;
}
```

Activity

Output:

```

.main {
  display: grid;
  grid-gap: 20px;
  grid-template-areas:
    "header header"
    "services services"
    "dogs cats"
    "contact contact";
}

```

PET'S CARE

Your Pet, Our Responsibility

OUR SERVICES

We are a group of animal lovers having certification in animal care. We provide various services like pet walking, grooming, feeding, playing with, and temporary shelter.

Pet Dog Services

Walking Services only on Monday, Wednesday and Friday - from 4pm to 10pm
Other Services are provided 5 days a week

Pet Cat Services

Walking Services only on Tuesday, Thursday and Saturday from 2pm to 8pm
Other Services are provided 5 days a week

Get in Touch

Phone Number - XXXXXXXXX
Email - petcare@gmail.com

Our layout is mostly finished, but not yet complete. We still have to design it in CSS.

- c. In the **.main** class, set the width to **80%**. Use the font-family property to make **'Lucida Sans'** the default font.

```

.main {
  display: grid;
  grid-gap: 20px;

  grid-template-areas:
    "header header"
    "services services"
    "dogs cats"
    "contact contact";
  width: 80%;
  font-family: Lucida Sans;
}

```

PET'S CARE

Your Pet, Our Responsibility

About Me

I'm an animal lover and pet owner. I have extensive experience walking, grooming, feeding, playing with, and caring for pets. I've been a volunteer at local animal shelters and have certification in animal care.

Pet Dog Timing

Monday, Wednesday and Friday - from 4pm to 6pm

Pet Cat Timing

Tuesday, Thursday and Saturday - from 4pm to 6pm

Get in Touch

Phone Number - XXXXXXXXX

- d. Right now, the main container is aligned to the left. To align it to the centre, set the **margin** property to **"auto"**.

```

"header header"
"about_me about_me"
"dogs cats"
"contact contact"
;

width: 85%;
font-family: 'Lucida Sans';
margin: auto;
}

```

PET'S CARE

Your Pet, Our Responsibility

About Me

I'm an animal lover and pet owner. I have extensive experience walking, grooming, feeding, playing with, and caring for pets. I've been a volunteer at local animal shelters and have certification in animal care.

Pet Dog Timing

Monday, Wednesday and Friday - from 4pm to 6pm

Pet Cat Timing

Tuesday, Thursday and Saturday - from 4pm to 6pm

Get in Touch

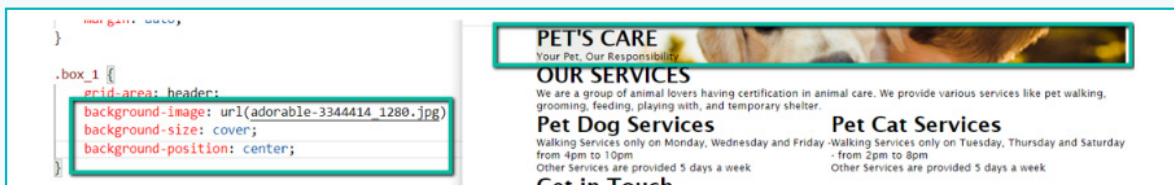
Phone Number - XXXXXXXXX
Email - petcare@gmail.com

Activity

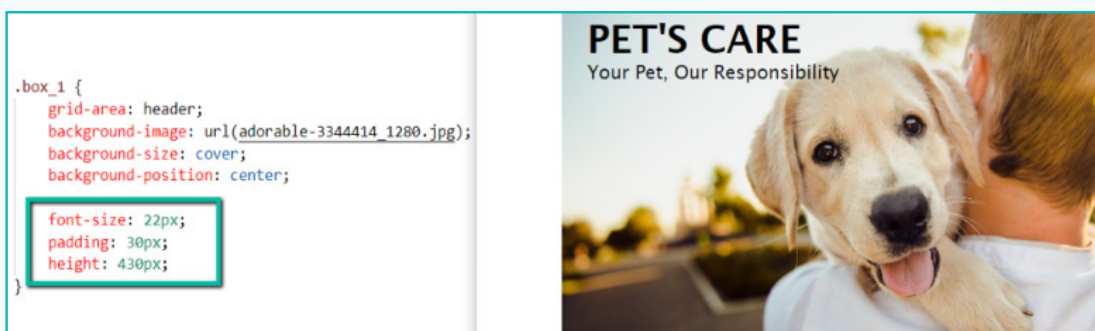
12 Design a header section

The header section is identified by the class name **.box_1**. Add a background image to our header container using the CSS property – “background-image”.

- a. **background-image**: A property used to add an image as background.
- b. **background-position**: A property which specifies a background image’s position. It can include values like **top**, **bottom**, **left**, **right** and **centre**.
- c. **background-size**: A property which specifies the size of a background image. It can include values like **cover**, **contain** and **auto**.



- d. At the moment, the content appears congested. Increase the height of the container using the CSS height property. We can also add some padding so that text does not stick to the edge of container.

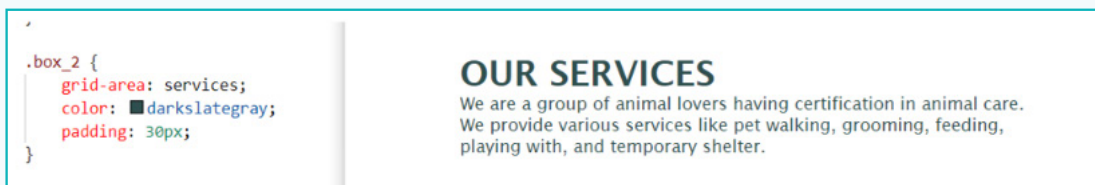


Activity

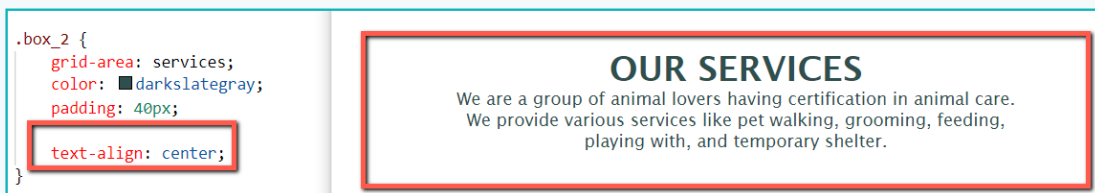
13 Design the Our Services section

The Our Services section is identified by the class name **.box_2**.

- a. Change the font colour using the **CSS color** property. Set some **padding** so that text fits better within the container.



- b. To make content appear at the centre of the page, use the CSS property **text-align: center**.



We can enhance the look of this section by adding a border around it. CSS provides a **border** property which can be used with any component.

For example: This image shows a red border around a text component:

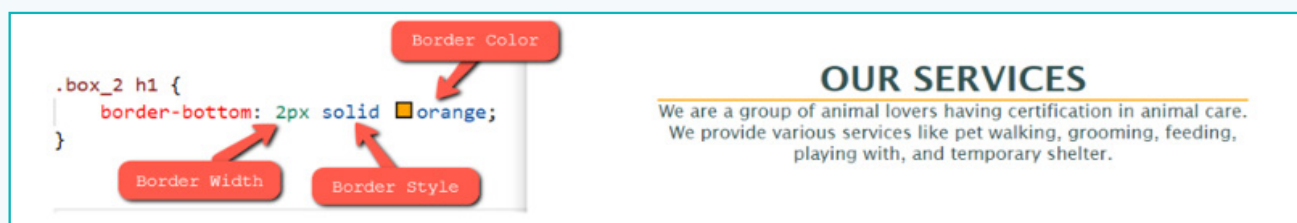
Border Sample

If we don't want a border on all four sides, we can use **border-left**, **border-right**, **border-bottom**, and **border-top** properties to apply a border on any one side.

- c. Apply a bottom border below the **<h1>** tag using the **border-bottom** property.

Note:

To specify the h1 tag inside the services section, we specify the class name (`box_2`) and then the `<h1>` tag.



Activity

- d. Add a gap between the border and the paragraph using the `padding-top` property. This will apply padding only on the top of the container. We can increase the gap between the lines of text by using the `line-height` property.

```
.box_2 p {  
  padding-top: 20px;  
  line-height: 35px;  
}
```

OUR SERVICES

We are a group of animal lovers having certification in animal care.

We provide various services like pet walking, grooming, feeding, playing with, and temporary shelter.

Our content looks much more attractive now. However, these properties are not mandatory. Their usage depends entirely on the person designing a webpage.

14 Design the section for Dogs and Cats Services

This section is identified by the class name `.box_3`.

- a. Centre align the content using `text-align: center`. Apply padding to it using the `padding` property and add a `border` to the right of it. The border will act as a single line separating two sections.

```
.box_3 {  
  grid-area: dogs;  
  border-right: 4px solid orange;  
  text-align: center;  
  padding: 15px;  
}
```

Pet Dog Services

Walking Services on Monday, Wednesday and Friday - from 4pm to 10pm.
Other Services are provided 5 days a week.

Pet Cat Services

Walking Services on Tuesday, Thursday and Saturday - from 2pm to 8pm.
Other Services are provided 5 days a week.

- b. Change the colour of the heading using the `color` property.

```
.box_3 h1 {  
  color: orange;  
}
```

Pet Dog Services

Walking Services on Monday, Wednesday and Friday - from 4pm to 10pm.
Other Services are provided 5 days a week.

Activity

- c. Add padding-top and increase the font size for the paragraph. This is optional and can be skipped if we don't want any spaces.

```

.box_3 p {
  padding-top: 15px;
  font-size: 18px;
}

```

Pet Dog Services

Walking Services on Monday, Wednesday and Friday - from 4pm to 10pm.

Other Services are provided 5 days a week.

- d. Repeat the same CSS properties for the **Cats Services section** (except the border property). This can be accessed with the class name **.box_4**.

```

.box_4 {
  grid-area: cats;
  text-align: center;
  padding: 15px;
}
.box_4 h1 {
  color: orange;
}
.box_4 p {
  padding-top: 15px;
  font-size: 18px;
}

```

Pet Dog Services

Walking Services on Monday, Wednesday and Friday - from 4pm to 10pm.

Other Services are provided 5 days a week.

Pet Cat Services

Walking Services on Tuesday, Thursday and Saturday - from 2pm to 8pm.

Other Services are provided 5 days a week.

15 Design the Contact/Footer section

This section is indicated with the class name **.box_5**.

- a. Use basic CSS to design this section. Feel free to add or remove design properties depending on the final output.

```

.box_5 {
  grid-area: contact;
  background-color: grey;
  padding: 15px;
  color: white;
  text-align: center;
  margin-top: 40px;
}

```

Get in Touch

Phone Number - XXXXXXXXXX
Email - petcare@gmail.com

- b. Change the heading colour to distinguish it from the other content.

```

.box_5 h1 {
  padding-bottom: 15px;
  color: orange;
}

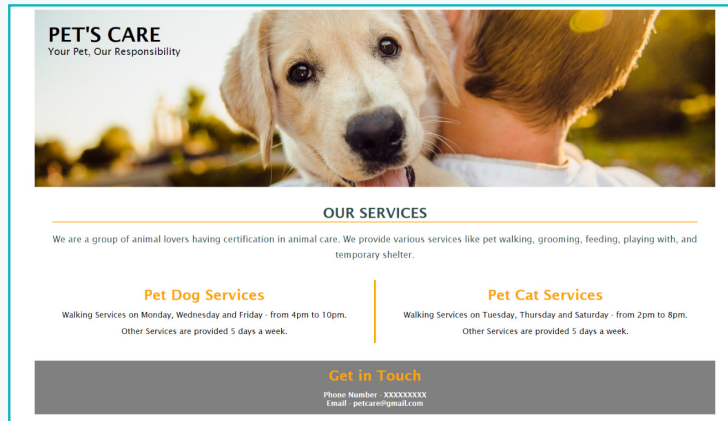
```

Get in Touch

Phone Number - XXXXXXXXXX
Email - petcare@gmail.com

Activity

Output:



We spent this lesson learning about website layouts. Now that we are familiar with layouts, we can decide which layouts work for different websites. We need some more information to design impactful websites. We need to know our audience, and their behaviour, needs and expectations. A well-designed layout clearly puts across a message in a convincing manner.

Exercise

Fill in the blanks

- 1 To access a class name, we use the _____ operator.
- 2 To create a Grid Container, we set the display property to _____.
- 3 We can assign a custom name to a grid item using the _____ property.
- 4 To arrange grid area items within a grid container, we use the _____ property.
- 5 We can align HTML components at the centre of a page by setting the margin property to _____.

Using this concept, a webpage will look beautiful and well organised. Always make sure to design or pick the layout that enhances your website.



Tech Flash

- **CSS Grid Layout** : A 2-dimensional layout system which allows us to divide a webpage into sections in a simple and easy manner.
- **Website Layout** : This defines a website's structure. It describes how the designer wants to arrange various HTML components or sections within a website.
- **Minimalist Web Design** : A website design concept which implements the following principles-
 - a. User-friendly Interface.
 - b. Limiting the colour scheme to only 2-3 colours.
 - c. Plenty of empty space in between components appearing on the site.
 - d. Fewer details like, colour transitions, shadows, textures, etc.